Synthesis and structure-activity relationships of new quinolone-type molecules against Trypanosoma brucei.

PMID: 22376072 | Indexed in PubMed

Human African trypanosomiasis (HAT) or sleeping sickness is caused by two subspecies of Trypanosoma brucei , Trypanosoma brucei gambiense , and Trypanosoma brucei rhodesiense and is one of Africa's old plagues. It causes a huge number of infections and cases of death per year because, apart from limited access to health services, only inefficient chemotherapy is available. Since it was reported that quinolones such as ciprofloxacin show antitrypanosomal activity, a novel quinolone-type library was synthesized and tested. The biological evaluation illustrated that 4-quinolones with a benzylamide function in position 3 and cyclic or acyclic amines in position 7 exhibit high antitrypanosomal activity. Structure-activity relationships (SAR) are established to identify essential structural elements. This analysis led to lead structure 29, which exhibits promising in vitro activity against T. b. brucei (IC(50) = 47 nM) and T. b. rhodesiense (IC(50) = 9 nM) combined with low cytotoxicity against macrophages J774.1. Screening for morphological changes of trypanosomes treated with compounds 19 and 29 suggested differences in the morphology of mitochondria of treated cells compared to those of untreated cells. Segregation of the kinetoplast is hampered in trypanosomes treated with these compounds; however, topoisomerase II is probably not the main drug target.

Identification of haptoglobin as a natural inhibitor of trypanocidal activity in human serum.

PMID: 7479764 | Indexed in PubMed

Trypanosomes are protozoan parasites of medical and veterinary importance. Trypanosoma brucei rhodesiense and Trypanosoma brucei gambiense infect humans, causing African sleeping sickness. However, Trypanosoma brucei brucei can only infect animals, causing the disease Nagana in cattle. Man is protected from this subspecies of trypanosomes by a toxic subtype of high density lipoproteins (HDLs) called the trypanosome lytic factor (TLF). The toxic molecule in TLF is believed to be the haptoglobin-related protein that when bound to hemoglobin kills the trypanosome via oxidative damage initiated by its peroxidase activity. The amount of lytic activity in serum varies widely between different individuals with up to a 60-fold difference in activity. In addition, an increase in the total amount of lytic activity occurs during the purification of TLF, suggesting that an inhibitor of TLF (ITLF) exists in human serum. We now show that the individual variation in trypanosome lytic activity in serum correlates to variations in the amount of ITLF. Immunoblots of ITLF probed with antiserum against haptoglobin recognize a 120-kDa protein, indicating that haptoglobin is present in partially purified ITLF. Haptoglobin involvement is further shown in that it inhibits TLF in a manner similar to ITLF. Using an anti-haptoglobin column to remove haptoglobin from ITLF, we show that the loss of haptoglobin coincides with the loss of inhibitor activity. Addition of purified haptoglobin restores inhibitor activity. This indicates that haptoglobin is the molecule responsible for inhibition and therefore causing the individual variation in serum lytic activity.

Ebsulfur is a benzisothiazolone cytocidal inhibitor targeting the trypanothione reductase of Trypanosoma brucei.

PMID: 23900839 | Indexed in PubMed

Trypanosoma brucei is the causing agent of African trypanosomiasis. These parasites possess a unique thiol redox system required for DNA synthesis and defense against oxidative stress. It includes trypanothione and trypanothione reductase (TryR) instead of the thioredoxin and glutaredoxin systems of mammalian hosts. Here, we show that the benzisothiazolone compound ebsulfur (EbS), a sulfur analogue of ebselen, is a potent inhibitor of T. brucei growth with a favorable selectivity index over mammalian cells. EbS inhibited the TryR activity and decreased non-protein thiol levels in cultured parasites. The inhibition of TryR by EbS was irreversible and NADPH-dependent. EbS formed a complex with TryR and caused oxidation and inactivation of the enzyme. EbS was more toxic for T. brucei than for Trypanosoma cruzi, probably due to lower levels of TryR and trypanothione in T. brucei. Furthermore, inhibition of TryR produced high intracellular reactive oxygen species. Hydrogen peroxide, known to be constitutively high in T. brucei, enhanced the EbS inhibition of TryR. The elevation of reactive oxygen species production in parasites caused by EbS induced a programmed cell death. Soluble EbS analogues were synthesized and cured T. brucei brucei infection in mice when used together with nifurtimox. Altogether, EbS and EbS analogues disrupt the trypanothione system, hampering the defense against oxidative stress. Thus, EbS is a promising lead for development of drugs against African trypanosomiasis.

Trans-sialidase from Trypanosoma brucei as a potential target for DNA vaccine development against African trypanosomiasis.

PMID: 19582478 | Indexed in PubMed

African trypanosomiasis (AT), also known as sleeping sickness in humans and Nagana in animals, is a disease caused by the protozoan parasite Trypanosoma brucei. AT is an extremely debilitating disease in human, cattle, and wild animals, and the treatment is difficult with frequent relapses. This work shows that BALB-c mice immunized intramuscularly with a single dose (100 microg) of a plasmid DNA encoding the 5'-terminal region of the trans-sialidase (nTSA) gene of T. brucei brucei are able to produce IgG antibodies that bind to the bloodstream form of T. brucei-protein extract and recognize the recombinant nTSA protein, expressed in Escherichia coli. Furthermore, this DNA vaccination process was able to protect 60% of mice submitted to a challenge assay with the infective form of T. brucei brucei parasites. These results demonstrate that a DNA vaccine coding for trans-sialidase from T. brucei is potentially useful in the prophylaxis of AT.

Structural and Functional Highlights of Vacuolar Soluble Protein 1 from Pathogen Trypanosoma brucei brucei.

PMID: 26494625 | Indexed in PubMed

Trypanosoma brucei (T. brucei) is responsible for the fatal human disease called African trypanosomiasis, or sleeping sickness. The causative parasite, Trypanosoma, encodes soluble versions of inorganic pyrophosphatases (PPase), also called vacuolar soluble proteins (VSPs), which are localized to its acidocalcisomes. The latter are acidic membrane-enclosed organelles rich in polyphosphate chains and divalent cations whose significance in these parasites remains unclear. We here report the crystal structure of T. brucei brucei acidocalcisomal PPases in a ternary complex with Mg(2+) and imidodiphosphate. The crystal structure reveals a novel structural architecture distinct from known class I

PPases in its tetrameric oligomeric state in which a fused EF hand domain arranges around the catalytic PPase domain. This unprecedented assembly evident from TbbVSP1 crystal structure is further confirmed by SAXS and TEM data. SAXS data suggest structural flexibility in EF hand domains indicative of conformational plasticity within TbbVSP1.

The role of copy number variants in the genetic architecture of common familial epilepsies.

PMID: 38101940 | Indexed in PubMed

Copy number variants (CNVs) contribute to genetic risk and genetic etiology of both rare and common epilepsies. Whereas many studies have explored the role of CNVs in sporadic or severe cases, fewer have been done in familial generalized and focal epilepsies. We analyzed exome sequence data from 267 multiplex families and 859 first-degree relative pairs with a diagnosis of genetic generalized epilepsies or nonacquired focal epilepsies to predict CNVs. Validation and segregation studies were performed using an orthogonal method when possible. We identified CNVs likely to contribute to epilepsy risk or etiology in the probands of 43 of 1116 (3.9%) families, including known recurrent CNVs (16p13.11 deletion, 15q13.3 deletion, 15q11.2 deletion, 16p11.2 duplication, 1q21.1 duplication, and 5-Mb duplication of 15q11q13). We also identified CNVs affecting monogenic epilepsy genes, including four families with CNVs disrupting the DEPDC5 gene, and a de novo deletion of HNRNPU in one affected individual from a multiplex family. Several large CNVs (>500 kb) of uncertain clinical significance were identified, including a deletion in 18q, a large duplication encompassing the SCN1A gene, and a 15q13.3 duplication (BP4-BP5). The overall CNV landscape in common familial epilepsies is similar to that of sporadic epilepsies, with large recurrent deletions at 15q11, 15q13, and 16p13 contributing in 2.5%-3% of families. CNVs that interrupt known epilepsy genes and rare, large CNVs were also identified. Multiple etiologies were found in a subset of families, emphasizing the importance of genetic testing for multiple affected family members. Rare CNVs found in a single proband remain difficult to interpret and require larger cohorts to confirm their potential role in disease. Overall, our work indicates that CNVs contribute to the complex genetic architecture of familial generalized and focal epilepsies, supporting the role for clinical testing in affected individuals.

A comprehensive analysis of clinical trials in pancreatic cancer: what is coming down the pike?

PMID: 33014285 | Indexed in PubMed

Pancreatic cancer is the most aggressive common cancer and is desperately in need of novel therapies. Unlike many other common cancers, there have been no new paradigm-changing therapies in the past 40 years beyond multi-agent chemotherapy. In this study, we perform the first comprehensive analysis of the current clinical trial landscape in pancreatic cancer to better understand the pipeline of new therapies. We queried https://clinicaltrials.gov/ for registered pancreatic cancer clinical trials. Studies were curated and categorized according to phase of study, clinical stage of the study population, type of the intervention under investigation, and biologic mechanism targeted by the therapy under study. As of May 18, 2019, there were 430 total active therapeutic interventional trials testing 590 interventions. The vast minority of trials (n = 37, 8.6%) are in phase III testing. 189 (31%) interventions are

immunotherapies, 69 (11%) target cell signaling pathways, 154 (26%) target cell cycle or DNA biology, and 35 (6%) target metabolic pathways. Of the late phase trials, only 14 are currently testing novel interventions. Rather, 23 phase III trials examine new ways to deliver existing FDA-approved drugs, procedures, or pain management. A large number of novel therapeutic strategies are currently under investigation. They include a broad range of therapies targeting diverse biologic processes. However, only a small number of novel therapies are in late-stage testing, suggesting that future progress is likely several years away, and dependent on the success of early-stage trials.

Biologics in gastrointestinal and pancreatic neuroendocrine tumors.

PMID: 28736633 | Indexed in PubMed

The development of biologic agents has ushered in a new era of precision medicine, opening the door to new therapeutic options designed to intelligently target cancer cells and their promoting factors, while leaving normal cells relatively unharmed. Biologics for the treatment of neuroendocrine tumors (NETs) have followed in the footsteps of regimens targeting pathways upregulated in other cancers, including the vascular endothelial growth factor (VEGF) and the mammalian target of rapamycin (mTOR). Through a number of clinical trials, the mTOR inhibitor everolimus and the receptor tyrosine kinase (RTK) inhibitor sunitinib were recently approved for NETs. Other biologics such as the VEGF-A inhibitor bevacizumab have also demonstrated promising clinical activity in NETs. Interestingly, though trials have demonstrated the efficacy of everolimus and sunitinib in extending progression-free survival (PFS) in NETs, objective response rates (RR) are uniformly low, indicating that the primary effect of these drugs is maintenance of stable disease. Due to the relatively indolent nature of the more common, well-differentiated variety of NETs, stable disease is often a reasonable goal for NET patients. Well-differentiated NETs have been shown to be poor responders to cytotoxic chemotherapy, underlining the important role of biologics in treating and managing NETs and their hormonal symptoms. Ongoing and future trials are investigating a wide variety of biologic compounds in NETs, including other RTK inhibitors, mTOR pathway inhibitors, and immune checkpoint inhibitors. Within this review, we will discuss major trials leading up to the FDA approval of everolimus and sunitinib for NETs, as well as other promising biologics currently under investigation in NET clinical trials.

Targeting Epidermal Growth Factor Receptor-Related Signaling Pathways in Pancreatic Cancer.

PMID: 26355547 | Indexed in PubMed

Pancreatic cancer is aggressive, chemoresistant, and characterized by complex and poorly understood molecular biology. The epidermal growth factor receptor (EGFR) pathway is frequently activated in pancreatic cancer; therefore, it is a rational target for new treatments. However, the EGFR tyrosine kinase inhibitor erlotinib is currently the only targeted therapy to demonstrate a very modest survival benefit when added to gemcitabine in the treatment of patients with advanced pancreatic cancer. There is no molecular biomarker to predict the outcome of erlotinib treatment, although rash may be predictive of improved survival; EGFR expression does not predict the biologic activity of anti-EGFR drugs in pancreatic cancer, and no EGFR mutations are identified as enabling the selection of patients likely to benefit from treatment. Here, we review clinical studies of EGFR-targeted therapies in combination with conventional cytotoxic regimens or multitargeted strategies in advanced pancreatic cancer, as well as research directed at molecules downstream of EGFR as alternatives or adjuncts to receptor targeting. Limitations of preclinical models, patient selection, and trial design, as well as the complex mechanisms underlying resistance to EGFR-targeted agents, are discussed. Future clinical trials must incorporate translational research end points to aid patient selection and circumvent resistance to EGFR inhibitors.

Comparative Study of 17-AAG and NVP-AUY922 in Pancreatic and Colorectal Cancer Cells: Are There Common Determinants of Sensitivity?

PMID: 25389454 | Indexed in PubMed

The use of heat shock protein 90 (Hsp90) inhibitors is an attractive antineoplastic therapy. We wanted to compare the effects of the benzoquinone 17-allylamino-17-demethoxygeldanamycin (17-AAG, tanespimycin) and the novel isoxazole resorcinol-based Hsp90 inhibitor NVP-AUY922 in a panel of pancreatic and colorectal carcinoma cell lines and in colorectal primary cultures derived from tumors excised to patients. PANC-1, CFPAC-1, and Caco-2 cells were intrinsically resistant to 17-AAG but sensitive to NVP-AUY922. Other cellular models were sensitive to both inhibitors. Human epidermal growth factor receptor receptors and their downstream signaling pathways were downregulated in susceptible cellular models, and concurrently, Hsp70 was induced. Intrinsic resistance to 17-AAG did not correlate with expression of ATP-binding cassette transporters involved in multidrug resistance. Some 17-AAG-resistant, NVP-AUY922-sensitive cell lines lacked quinone oxidoreductase 1 (NQO1) enzyme and activity. However, colorectal LoVo cells still responded to both drugs in spite of having undetectable levels and activity of NQO1. Pharmacological and biologic inhibition of NQO1 did not confer resistance to 17-AAG in sensitive cell lines. Therefore, even though 17-AAG sensitivity is related to NQO1 protein levels and enzymatic activity, the absence of NQO1 does not necessarily convey resistance to 17-AAG in these cellular models. Moreover, NVP-AUY922 does not require NQO1 for its action and is a more potent inhibitor than 17-AAG in these cells. More importantly, we show in this report that NVP-AUY922 potentiates the inhibitory effects of chemotherapeutic agents, such as gemcitabine or oxaliplatin, and other drugs that are currently being evaluated in clinical trials as antitumor agents.